Design Objectives for the SIGMA Computer

A. Purpose

The primary purpose of this computer system is to fulfill our contractural obligations with the Atomic Energy Commission. A secondary, but perhaps more fundamental purpose is to provide the free world with the best general purpose computer which the latest advances in solid state components can practically produce. Care must be taken in the planning to insure consistantly high performance by all parts of the computer system. The SIGMA computer must also have a high standard for reliability and ease of maintenance in order to make its high speed effective on long and difficult problems.

B. Market

1. The SIGMA computer will be most applicable to those problems, largely scientific in nature, requiring a large amount of computing time with relatively small amounts of original input and final output. There will be large amounts of high speed communication with auxiliary storage, however.

2. The immediate market will be in the Atomic Energy field in the design and simulation of nuclear weapons and reactors. A field of growing importance will be in guided missle system design and evaluation. Later applications in the fields of chemical kinetics, economic "game theory", military operations research, real-time simulation and control of our air defense network, and many problems of the coming "space age" etc., are certain to develop when a computer capable of actually solving such problems becomes available.

3. The SIGMA arithmetic speeds and data handling capacities should be aimed at solving the problems already in existance which are not solvable on present computer systems. With advanced computer-assisted coding systems and auto-monitoring techniques, the SIGMA system should also find a market in those installations which have large quantities of medium-size problems such as the engine design simulation, flutter analysis, wind tunnel data reduction, and control system studies in the aircraft industry. It should be noted that the existance of a computer capable of handling large problems efficiently will encourage the integration of many smaller problems done separately today into single, large problems.

4. The rental objective for a SIGMA system ~~consisting~~ exclusive of tapes and adapters is as follows: ~~of a central processing unit, a console, four units of two microsecond memory (16,384 words per unit), two units of half-microsecond memory (1024 words per unit), a high speed disk file with its exchange, a regular exchange with suitable tapes and other input-output equipment should be $200,000, per month.~~ see other paper $170,000 to $200,000

5. There should be a market for 20 to 30 SIGMA systems over a five year period.

C. Fundamental Characteristics

1. The performance of the SIGMA computer on typical applications in the field of nuclear weapon or reactor design should be 100 times faster than the 704 or 709 on the internal computing portions of the problems. The ~~input-output~~ auxiliary storage required for such problems must be fast enough and handled in such a way so as to not reduce this performance materially.

2. The computer will be capable of ~~performing~~ performing arithmetic in fixed point, and in normalized and unnormalized floating point. Special provisions will be made to facilitate multiple-precision arithmetic.

part of B.4.

Sigma Main Frame	75 000	90 000
4 units of 2 ypsec memory	44,000	52,800
2 units of 0.6 ypsec memory	22,000	26,400
1 150 ypsm remote and adapter	1800	2160
1 250 ypsm punch "	1600	1920
1 1000 ypsm reader "	1200	1440
1 Duplex Station and adapter	200	240
1 High speed Exchange	8,000	9,600
1 Disc adapter	3500	4200
1 disc unit	3500	4200
1 exchange	8,000	9,600
	168 800	202,560
	20260	

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3. The data word size shall be 64 bits and sign for data words plus the 8 bits necessary for error detection and correction. Most instructions will be 32 bits in length, with some more complicated ones requiring 64 bits. Floating points words will contain a 48 bit mantissa plus mantissa sign, an eleven bit exponent, plus exponent sign, and three tag bits.

4. The nominal arithmetic speeds should be as follows:

Fixed Addition	0.2 microseconds
Floating Addition	0.6 microseconds
Fixed or Floating Multiply	1.2 microseconds
Fixed or Floating Divide	1.8 microseconds
Load or Store	0.2 microseconds

The contract with the AEC allows a 25% imprecision in the above times. It also states that the time required to transfer data from the memory registers to the arithmetic unit will normally add no more than 0.2 microseconds to the above times.

5. There shall be at least four units of core memory, each containing 16,384 words having a full read-write cycle time of 2.0 to 2.2 microseconds. These units will be used mainly for data storage and will be capable of operating concurrently and asynchronously with respect to the rest of the system.

6. There shall be at least two units of core memory, each containing 1,024 words, with a full cycle time of 0.6 to 0.7 microseconds. These units will be used mainly for instruction storage and will also be capable of operating concurrently and asynchronously with respect to the rest of the system.

7. There shall be at least sixteen words of storage with a 0.2 microsecond cycle time which will be mainly used for index quantities and temporary storage of intermediate results.

8. To quarantee a high degree of efficiency in the arithmetic unit consecutive memory addresses will be distributed among the various storage units and a multiplexing device shall be provided to pre-reference memory so that as much of the time as is possible the data will be available without causing a delay in the arithmetic unit.

9. Automatic checking will be provided throughout the system. The checking will detect all single errors and most multiple errors. Parts of the system will be designed to detect single, double, and most multiple errors, as well as to correct all single errors. During arithmetic operations the arithmetic unit will repeat its operation if an error is detected. The automatic checking error correction, and marginal checking

mechanisms will be specifically designed to reveal the nature and location of machine faults where possible.

10. An editing system will be provided to arrange input data for use by the computer and to convert computer results into the form required by output devices. The conversion of data from one radix to another or from one machine code to another must be done quickly and conveniently.

11. The computer will include new and powerful logical instructions including all of the sixteen logical connectives which may be applied under program control upon any addressable 64 bit word in the computer. The instruction set and formats must be chosen to insure that the SIGMA system has the greatest possible effectiveness on large and complex programs. The instruction set must also be in harmony with the requirements of machine-assisted coding.

D. Minimum Performance Criteria

The minimum performance criteria are described in a memo dated December 30, 1957, by D. A. Jameson, E. H. Carlstrom, and H. G. Kolsky.

E. Installation Considerations

1. The AEC SIGMA system will be of such a size and layout that it can be installed in a room 54 feet long by 54 feet wide with a ceiling eight feet high. Raised flooring should not be necessary.

2. The power and air conditioning requirements will not be more than the requirements of for three 704 systems. The solid state components should allow a much lower requirements than these in present practice.

3. The noise level at the operator's console will be as low as is reasonable possible.